

Pablo GROBAS ILLOBRE

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Computational Chemist

Scientific Software Developer

PROFILE SUMMARY

Computational chemist and scientific software developer with 6+ years' experience in computational modeling, QM/MM, and HPC workflows. Skilled in C++/Fortran/Python programming with a proven track record in scientific software development and international collaborations. Currently transitioning toward the drug discovery industry, applying physics-based ML and high-performance computing to molecular modeling challenges.

CONTACT DETAILS

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LANGUAGES

- Spanish (native)
- Galician (native)
- English (fluent)
- Italian (fluent)
- French (fluent)

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Fortran, Matlab
- **Python Libraries:** RDKit, TensorFlow, PyTorch, Scikit-learn, Keras, NumPy, SciPy, Matplotlib
- **Fortran Development:** OpenMP parallelization, CMake compilation
- **Development Tools:** Git, Visual Studio Code, Vim
- **Operating Systems and Scripting:** Proficient in Linux and Bash/shell scripting
- **Interactive Environments:** Jupyter, Google Colab
- **Scientific Software:** Amsterdam Modeling Suite, Gaussian
- **High Performance Computing (HPC) environments**

EXPERIENCE AND TRAINING

Postdoctoral Researcher,

Scuola Normale Superiore | 02/2025 - Present | Pisa, Italy

Currently developing **QM/MM quantum chemistry software** in **C++** and **Fortran** in collaboration with the *Software for Chemistry & Materials* company. This work is complemented by **Python programming** for **machine learning, data analysis, statistics, and visualization**.

My research integrates intensive **software development** to investigate:

- **QM/MM Surface-Enhanced Spectroscopies:** Fluorescence, Raman scattering, Raman optical activity.
- **Plasmonic Materials:** graphene & metal nanoparticles.
- **Plasmon Mediated Electronic Energy Transfer.**

Ph.D. (*cum laude*) in Methods and Models for Molecular Sciences,

Scuola Normale Superiore | 11/2020 – 01/2025 | Pisa, Italy

Title: Modeling Atomistic Nanoplasmonics: Classical and Hybrid Quantum Mechanical/Classical Schemes

Supervisor: Chiara Cappelli

M.Sc. in Theoretical Chemistry and Computational Modeling,

Université Paul Sabatier | 09/2018 – 08/2020 | Toulouse, France

M.Sc. Thesis, University of Trieste

11/2019 – 08/2020 | Trieste, Italy

Fortran software development for quantum chemistry

Visiting Student, Autonomous University of Madrid

09/2019 - 10/2019 | Madrid, Spain

M.Sc. Internship, Okayama University

04/2019 - 06/2019 | Okayama, Japan

Molecular dynamics simulations of ions in solution.

Research Assistant, Instituto de Tecnología Química

07/2019 – 08/2019 | Valencia, Spain

Molecular dynamics for sugar separation in the food industry.

Supervisor: Germán Ignacio Sastre Navarro

B.Sc. in Chemistry, University of A Coruña

09/2014 – 07/2018 | A Coruña, Spain

Erasmus Exchange, University of Oslo | 08/2016 - 12/2016

RESEARCH PROJECTS

GEMS - General Embedding Models for Spectroscopy, ERC Consolidator Grant

Scuola Normale Superiore | 2020 - 2025 | PI: Chiara Cappelli

- **Python** programming (in-house codes) for **data analysis, manipulation, and visualization** (e.g., see examples of Python-driven figures in T. Giovannini *et al.*, *ACS Photonics*, 2022, 9, 3025).
- Currently developing a **machine learning** pipeline in **Python** to study and simulate graphene samples.
- Quantum chemistry **software development** in **Fortran** (within the Amsterdam Modeling Suite and in-house codes) for modeling plasmonic materials, **QM/MM** surface-enhanced Raman scattering, and Raman optical activity.
- Intensive use of **HPC** infrastructures to streamline large-scale **data production** workflows.

FARE - “Framework per l’attrazione e il rafforzamento delle eccellenze per la ricerca in Italia”

Scuola Normale Superiore | 2020 - 2025 | PI: Chiara Cappelli

- **Python** programming (in-house codes) for **data analysis, manipulation, and visualization** (e.g., see generated Python-driven figures in P. Grobas Illobre, *et al.*, *Nanoscale Adv.*, 2024, 6, 3410).
- Quantum chemistry **QM/MM software development** in **C++** and **Fortran** (within the Amsterdam Modeling Suite and in-house codes) for modeling surface-enhanced fluorescence, and plasmon-mediated electronic energy transfer.
- Intensive use of **HPC** infrastructures to automate and streamline **data production** workflows.

SELECTED PUBLICATIONS

- **P. Grobas Illobre**, P. Lafiosca, T. Guidone, F. Mazza, T. Giovannini, C. Cappelli, "Multiscale Modeling of Surface Enhanced Fluorescence", *Nanoscale Adv.*, 2024, **6**, 3410.
- **P. Grobas Illobre**, P. Lafiosca, L. Bonatti, T. Giovannini, C. Cappelli, "Mixed atomistic-implicit quantum/classical approach to molecular nanoplasmonics", *J. Chem. Phys.*, 2025, **162**, 044103.

SUPERVISION & TEACHING

B.Sc. Thesis: “Fluorescence Response of Chromophores Near Plasmonic Nanoparticles: Atomistic vs. Continuum Approaches” – Teresa Guidone (2023), *Scuola Normale Superiore*

I supervised B.Sc. student Teresa Guidone for her thesis project, which included:

- Reviewing scientific literature in quantum chemistry, specifically on surface-enhanced fluorescence.
- Teaching Fortran95 (including parallelization) and Python programming.
- Guiding Fortran95 algorithm design, implementation, and debugging in the Amsterdam Modeling Suite.
- Leading Python data analysis and manipulation.
- Instructing in the use of Linux and High-Performance Computing (HPC) environments.

Course Teaching: Advanced Topics in Quantum Chemistry (2024), *Scuola Normale Superiore*

I delivered practical sessions in quantum chemistry for Ph.D. and master’s students, which included:

- Teaching various energy decomposition methods within quantum chemistry.
- Organizing hands-on sessions with the GAMESS quantum chemistry software.
- Training in Linux and HPC environment usage.
- Providing guidance on data generation and analysis.

AWARDS

- PhD awarded with honors (*cum laude*)
- Scuola Normale Superiore PhD Scholarship (2020-2025).
- Two NanoX Research Scholarships (2019 & 2020).
- Erasmus Internship Fellowship (2019-2020).
- Consejo Superior de Investigaciones Científicas (CSIC) – Jae Intro Fellowship (2019).
- Extraordinary Prize for achieving the best academic record in the Bachelor of Chemistry (2018).
- Erasmus+ scholarship (2016).